

Orhan Kaplan, Ph.D.

Senior Data Scientist | Gaziantep, Turkey | orhankaplan@gantep.edu.tr | +90 5464619339 | <https://nalpako2027.github.io/> | <https://github.com/nalpako2027?tab=repositories> | <https://scholar.google.com/citations?user=1q5Kk4UAAAJ&hl=en>

PROFESSIONAL SUMMARY

Senior Data Scientist specializing in causal inference and predictive behavioral/psychological modeling, with nearly 10 years of R&D experience including a graduate research assistantship at the University of Florida, lecturer and researcher at Gaziantep University and a record of peer-reviewed publications. Skilled at turning messy, large-scale user data into rigorous program and impact evaluations — designing studies, building statistical frameworks that isolate true feature impact where standard A/B testing is inapplicable, and delivering impactful findings into clear decision-making insights with sophisticated visuals for stakeholders across industries.

TECHNICAL SKILLS

Data Science/Applied Research: Hypothesis Testing, Experimental/Quasi-Experimental Observational Studies Design, Structural Equation Modeling, Propensity Score Matching, Entropy Balancing, Fixed-Effects Modeling, Machine Learning (Multiple Regression, Lasso & Ridge Regression, Logistic Regression, kNN Regression/Classification, Decision-Tree ensembles), Multiple imputation (MI, MICE), and Qualitative Research.

Programming & Development: Languages & Tools (Python, SPSS, Stata, MPlus, Git, SQL (SQLiteStudio, MySQL)), **Machine Learning** (scikit-learn, cross-validation, classification, Random Forest, XGBoost), and **Data & Visualization** (pandas, NumPy, Stata, matplotlib/seaborn, Streamlit).

Communication & Collaboration: Strong in rigorous academic/technical writing, team collaboration, supporting a productive and inclusive environment.

Languages: Fluent in English and Turkish.

PROFESSIONAL EXPERIENCE

Lecturer & Researcher | Gaziantep University, *April 2019 – Present*

- Designed/analyzed data for research projects using SPSS/Stata/Mplus/Dedoose/Python
- Peer-reviewed publications on research on psychological, educational, behavioral modeling
- Taught courses on research methodology at graduate level (delivered in English) and undergraduate level (mathematics, teaching methods and vocational English).
- Collaborated on research projects, social awareness projects, and built a Mathematics Library

Graduate/OPS Research Assistant | Higher Education Department, University of Florida, *October 2016 – May 2018*

- Managed and cleaned large-scale datasets
- Screened datasets and conducted statistical analyses using Stata (Regression-Discontinuity Design, Fixed-effects Modeling, Logistic Regression)
- Screened state-level educational policies and literature review

Mathematics Instructor | Upward Bound Program, University of Florida, *January 2012 – May 2012*

- Instructed high school students in ACT-oriented geometry.
- College preparatory program for students in poverty.

Mathematics & Science Teacher | Batman, Turkey, *September 2006 – February 2008*

- İkiztepe Middle School (2006-2008) and Şehit Öğretmenler Middle School (2007-2008)

SELECTED PROJECTS

Warehouse Closure Analysis: A SQL-Based Multi-Criteria Decision Framework: A SQL case study using the Mint Classic sample database to determine which warehouse, if any, should be closed, using a two-level weighted scoring model built

entirely from queryable data using MySQL Workbench. The decision-making framework visualized with an interactive dashboard using Streamlit and deployed on Streamlit Community Cloud.

MICE-PMM Diagnostics Toolkit: Open-source Python package implementing MICE with Predictive Mean Matching (Rubin, 1987; van Buuren, 2018) for complex survey data, with a diagnostics suite (Rubin's Rules, FMI, RVI, RE, Von Hippel's recommended-M, convergence plots) not previously available together in existing open-source tools.

Telco Customer Churn Analysis & Predictive Modeling Pipeline: End-to-end pipeline identifying high-risk churn segments and simulating ROI of retention strategies using logistic regression and KNN classification, with custom data quality checks and multi-model ROC evaluation.

Sci-Fi Movie Recommendation System Using KNN Collaborative Filtering: Built a content recommendation engine using distance-based KNN regression, optimized with Manhattan distance and validated via a 75/25 train-test split.

Applications of AI in Mathematics Instruction | TÜBİTAK Grant 222B158 (2023): Instructed mathematics educators on integrating AI-driven tools into instructional design, as part of a nationally funded technological innovation initiative for math education.

EDUCATION

Ph.D. in Curriculum & Instruction , University of Florida, FL, USA	2018
Specialization in Mathematics Education and Research and Evaluation Methodology	
M.S.E. in Mathematics , Arkansas State University, AR, USA	2011
Thesis on <i>Stone-Weierstrass Approximation Theorem</i>	
Studying Abroad Scholarship , Ministry of National Education, Turkey	October 2007
A selective and comprehensive scholarship covering education for Master's and PhD degrees	

CERTIFICATIONS

SQL for Data Analysis and Business Insights , Coursera,	August 2026
Machine Learning and AI , HarvardX,	June 2026
Machine Learning and AI Foundations: Causal Inference and Modeling , LinkedIn Learning,	June 2026
Learning Git and GitHub , LinkedIn Learning,	June 2026
Introduction to Data Science with Python , HarvardX,	May 2026

SELECTED PUBLICATIONS & PRESENTATIONS

Kaplan, O. (2026). Cross-cultural control–value mechanisms on the detrimental effect of bullying on mathematics anxiety. *Behavioral Sciences*, 16(1), Article 3. <https://doi.org/10.3390/bs16010003>

Aydın Gürler, S. & Kaplan, O. (2026). The effect of school bullying on science achievement and cognitive appraisals. *Problems of Education in the 21st Century*, 84(1), 7–26. <https://doi.org/10.33225/pec/26.84.07>

Kaplan, O., Kacar, T. & Simsek, N. (2025). The effect of bullying on epistemic emotions using Propensity Score Matching: Examining cross-national differences. The paper was presented at 2025 European Conference on Educational Research, Belgrade, Serbia.

Kaplan, O., Kara, M., & Kaçar, T. (2025). The effect of digital technology integration on students’ creative thinking skills: PISA evidence using propensity score matching. *Problems of Education in the 21st Century*, 83(3), 403–423. <https://doi.org/10.33225/pec/25.83.403>

Aydın Gürler, S. & Kaplan, O. (2023). Attitudes towards STEAM, critical thinking disposition and decision-making skills: Mediation and gender moderation. *International Journal of Contemporary Educational Research*, 10(1), 210-223. <https://doi.org/10.33200/ijcer.1272051>

Full publication list available at <https://orcid.org/0000-0002-4306-2266> or <https://nalpako2027.github.io/about/>